CURRICULUM VITAE

Nejc Blaznik

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Profile

Experimental physicist with a strong background in **quantum mechanics, optics** and **data-driven imaging**. Strong blend of **laboratory expertise, software engineering** and **data analysis**. Built new imaging methods, custom reconstruction pipelines and robust analysis tools for complex experiments. Interested in problems where physics, computation and instrumentation meet.

Experience

AtomQL May 2026-present

- Part of the team building a strontium apparatus to trap and neutralise Sr^+ ions in optical tweezers, contributing across vacuum, laser systems, and optics.
- Redesigned and rebuilt the group's data-management and analysis structure.

Ultracold Quantum Gases Laboratory at Utrecht University 2020-2026

- Operated and maintained a cold-atom experiment with sodium BECs.
- Developed and implemented new techniques for spin-resolved off-axis holography.
- Designed PCA-based phase reconstruction and background-inpainting pipelines *(manuscript in preparation)*.
- Created a Python+PyQt5 analysis suite for controlling the experiment and analysing spin dynamics, spin segmentation, and parameter correlation across experiments.

Microsoft Quantum Lab at QuTech 2019-2020

- Performed gate-based reflectometry analysis and noise characterization of superconducting charge islands.
- Built Python workflows for signal extraction and device characterization.

SRON (Netherlands Institute for Space Research) 2018

- Analysed high-resolution X-ray spectra of AGN 3C 273 using SPEX spectral fitting program.
- Worked extensively in Unix environments and command-line tools.
- Produced quantitative interpretation of absorption and emission features.

HiSPARC Cosmic Ray Detection Project 2016

- Studied weather effects on cosmic-ray detection.
- Focused on clear data presentation and statistical interpretation.

IB Academy 2016-present

- Tutored students in physics and mathematics, including intensive revision session.
- Created a study guide.
- Developing IB Academy Website, scheduling system and invoicing tools.

Zapik 2011-2020

- Science Correspondent for a Slovenian Magazine.

Education

Utrecht University | PhD in Experimental Physics
2020-2026

PhD Thesis: *Spin-resolved Holography in Ultracold Sodium Gases*.

Supervisors: prof.dr. Peter van der Straten, dr. Dries van Oosten

PhD thesis defended: March 2026

Utrecht University | MSc in Experimental Physics
2018-2020

MSc Thesis: *Investigating Quasiparticle Poisoning in Floating Semiconductor-Superconductor Hybrid Devices With Gate-Based Reflectometry Sensing*.

Supervisors: Prof. dr. Peter van der Straten, Prof. Dr. Ir. Leo. P. Kouwenhoven

Young Innovators honours programme

University College Utrecht | Liberal Arts and Sciences
2015-2018

BSc Thesis: *Understanding the X-ray absorption and emission in archetypal radio-loud AGN 3C 273 with XMM-Newton RGS*.

Supervisor: dr. Missagh Mehdipour

Major: Physics & Mathematics, **Minor**: Performing Studies

Graduated with Honours

Kranj High School (IB)
2013-2015

HL: Math, Physics, English, Slovene, **SL**: Chemistry, Philosophy

Extended Essay in Physics

Awards & activities

Conferences & posters

Utrecht University Open Day (2021)

AMO Conference (2022, 2023)

Summer Physics School Bad Honnef (2024)

Leadership & activities

Gold, Silver and Bronze The Duke of Edinburgh Award

Former National U18 Chess Champion

Leadership roles in UCSA committees, managing events and budgets

Founded Hermes, the largest sports committee at UCU

Technical skills

Programming & data

Python (NumPy, SciPy, Matplotlib, scikit-image, scikit-learn, OpenCV) · PyTorch (GPU ops, optimisation) · SVD/PCA pipelines · noise modeling · image reconstruction · Jupyter · Linux/WSL · shell scripting

Software engineering

PyQt5 (front-end + back-end) · Arduino / ESP8266 control · Git/GitHub, SQL · HTML, CSS, JS · React, Vite, Tailwind, Framer Motion

Experimental systems

Digital Holography · Numerical phase reconstruction · Optics and imaging · RF control, magnetic trapping, noise diagnostics · Electronics · Oscilloscopes, function generators, PID controllers

Tools & creative

Latex · Photoshop · Inkscape · Blender · Music theory · Mathematica · MS Excel and Google Sheets scripting

Languages

Slovenian	Native	**English**	Fluent
Serbo-Croatian	Proficient	**Dutch**	Conversational

Selected publications & patents

Blaznik, N., et al. (2024). Imaging spinor Bose gases using off-axis holography. Phys. Rev. A, 110, 023303.

Blaznik, N., et al. (2026). Coherent Spin Control and Zeeman-State Reconstruction in a Spin-1 Sodium Gas (in preparation).

Blaznik, N., Mosk, A.P. (2026). Background correction method for noise reduction in holography (patent submitted).

Blaznik, N. (2020) Investigating Quasiparticle Poisoning in Floating Hybrid Double Dot Devices Using Gate-Based Sensing, Master's thesis

Blaznik, N., et al. (2025). Polarization-sensitive imaging in magnetic environments. SciPost Physics Core (Accepted).

Blaznik, N., et al. (2026). Investigating Spin Diffusion in Spinor Bose Gases (in preparation).

De Jong, D., Waardenburg, D., **Blaznik, N.**, Han, L., Malinowski, F., Prosko, C., Van Veen, J,, Krogstrup, P., Kouwenhoven, L. P. and Pfaff, W. (2020). Dispersive sensing of electron tunneling between quantum dots in proximitized InAs nanowires. In APS March Meeting